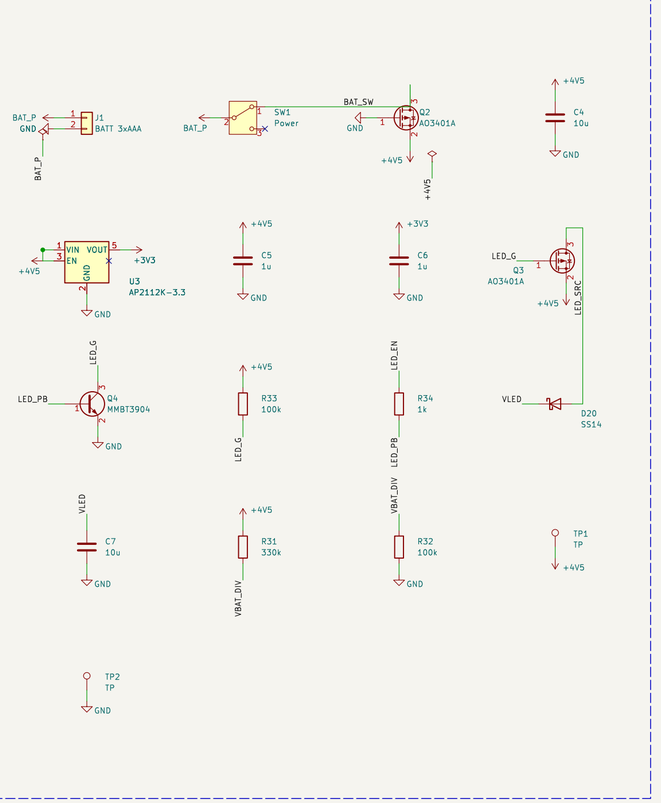
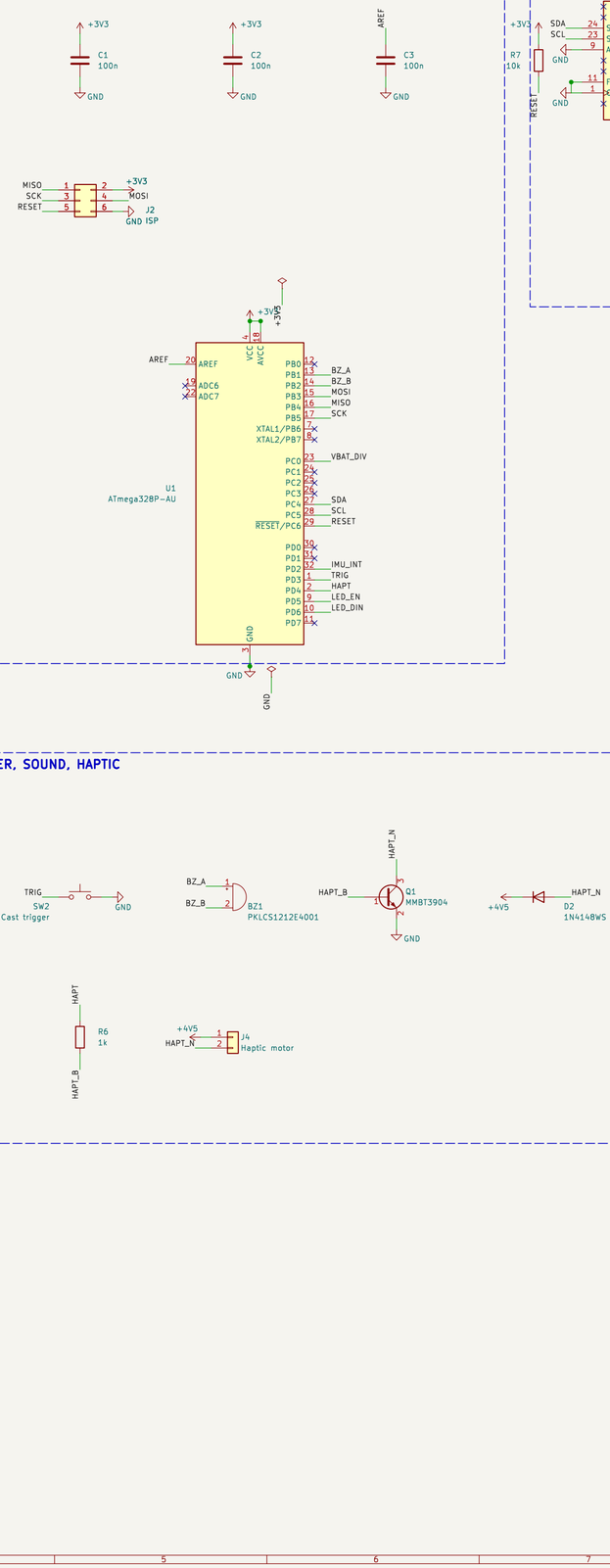
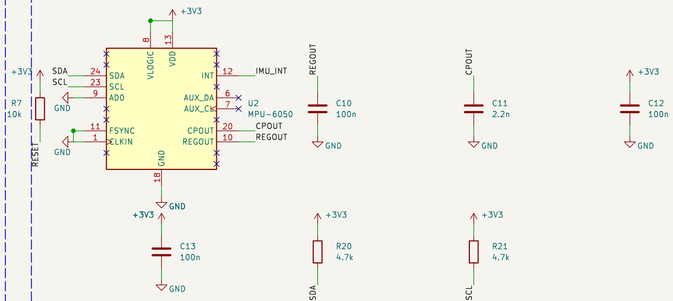
POWER: 3xAAA, switch, reverse-polarity PMOS, 3.3 V LDO, LED supply diode, battery divider



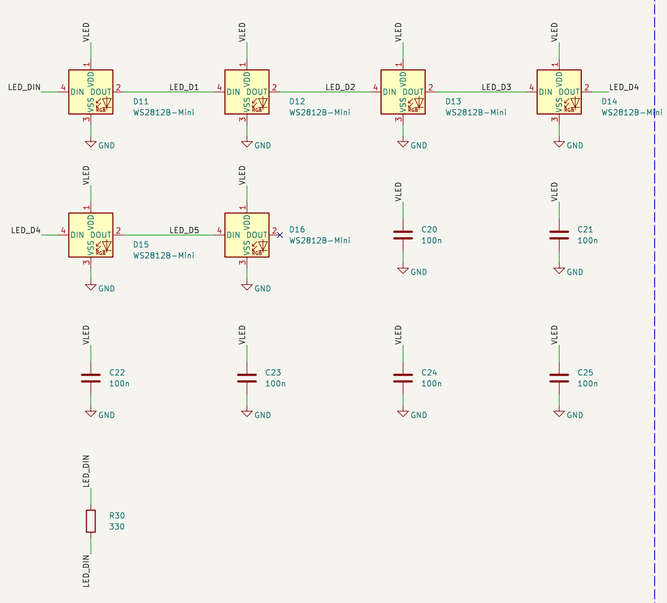
MCU: ATmega328P-AU, 3.3 V, 8 MHz internal



MOTION: MPU-6050 6-axis IMU on I2C



LIGHT: six WS2812B-Mini pixels (daisy chain), each with 100 nF



TRIGGER, SOUND, HAPTIC

ASTRADAND: hold the trigger, draw a shape in the air, release. The 6-axis IMU is read over I2C at 50 Hz; the ATmega matches the drawn path against six spell templates. Piezo is driven differentially from OC1A/OC1B (D9/D10). LEDs are fed from +4V5 through a high-side switch (Q3, driven by Q4 from D5) and a Schottky (D20), so the pixels draw nothing in sleep so that their 0.7 x VDD data threshold stays below the 3.3 V logic level; the firmware caps the pixel current.

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheet: /
File: astradand_main.kicad_sch

Title: *ASTRADAND gesture wand — main board*

Size: A2	Date: 2026-11-01	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1